2026 Spring Conference of the Korean Society of Medical Informatics
2026 대한의료정보학회 춘계학술대회
2026. 6. 24.수 - 26.목
청주오스코

BERT Models for Korean Clinical NLP: Comparing Domain-Adaptive Pretraining and From-Scratch Pretraining Approaches



박영준
가천대학교

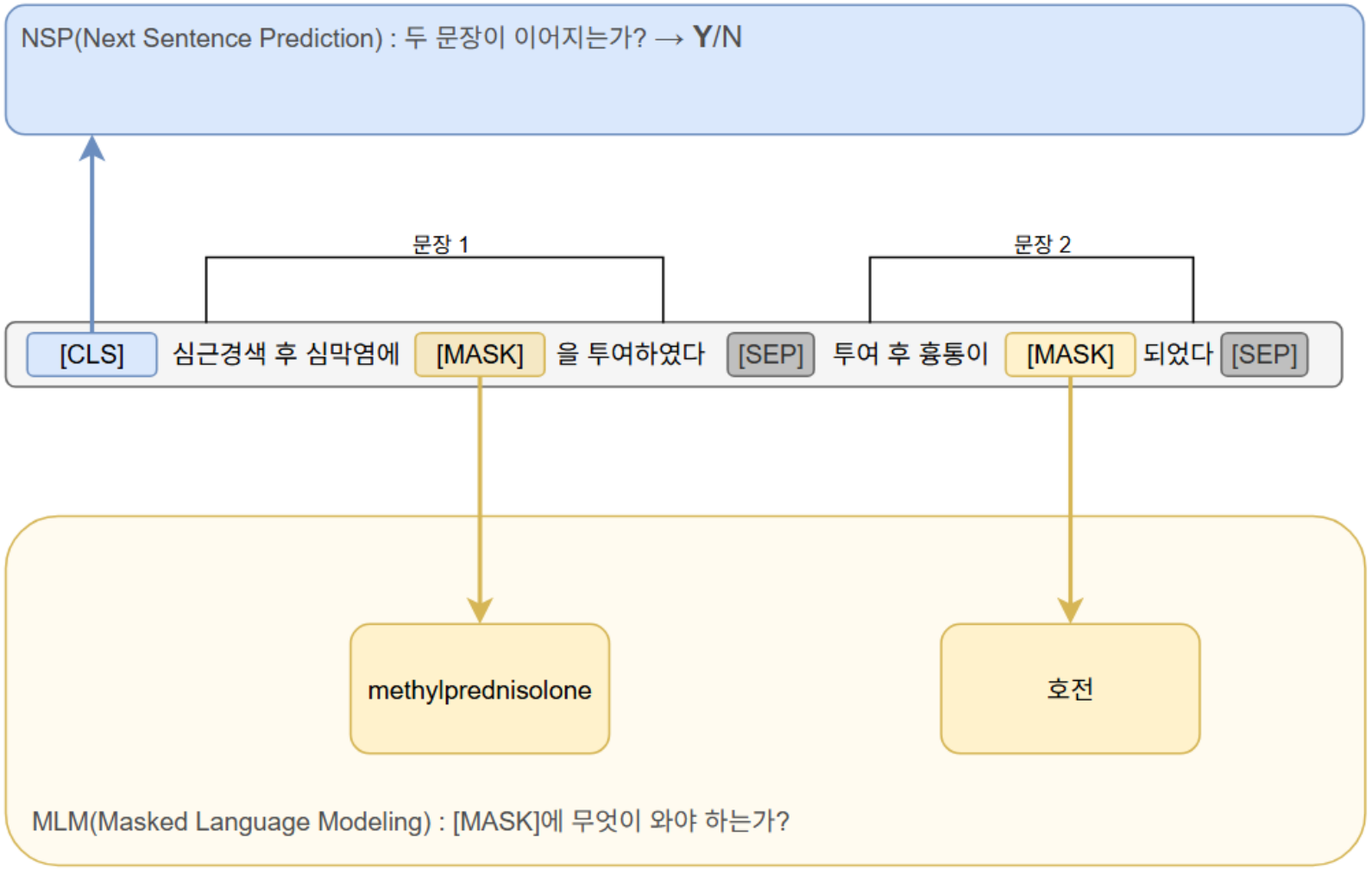
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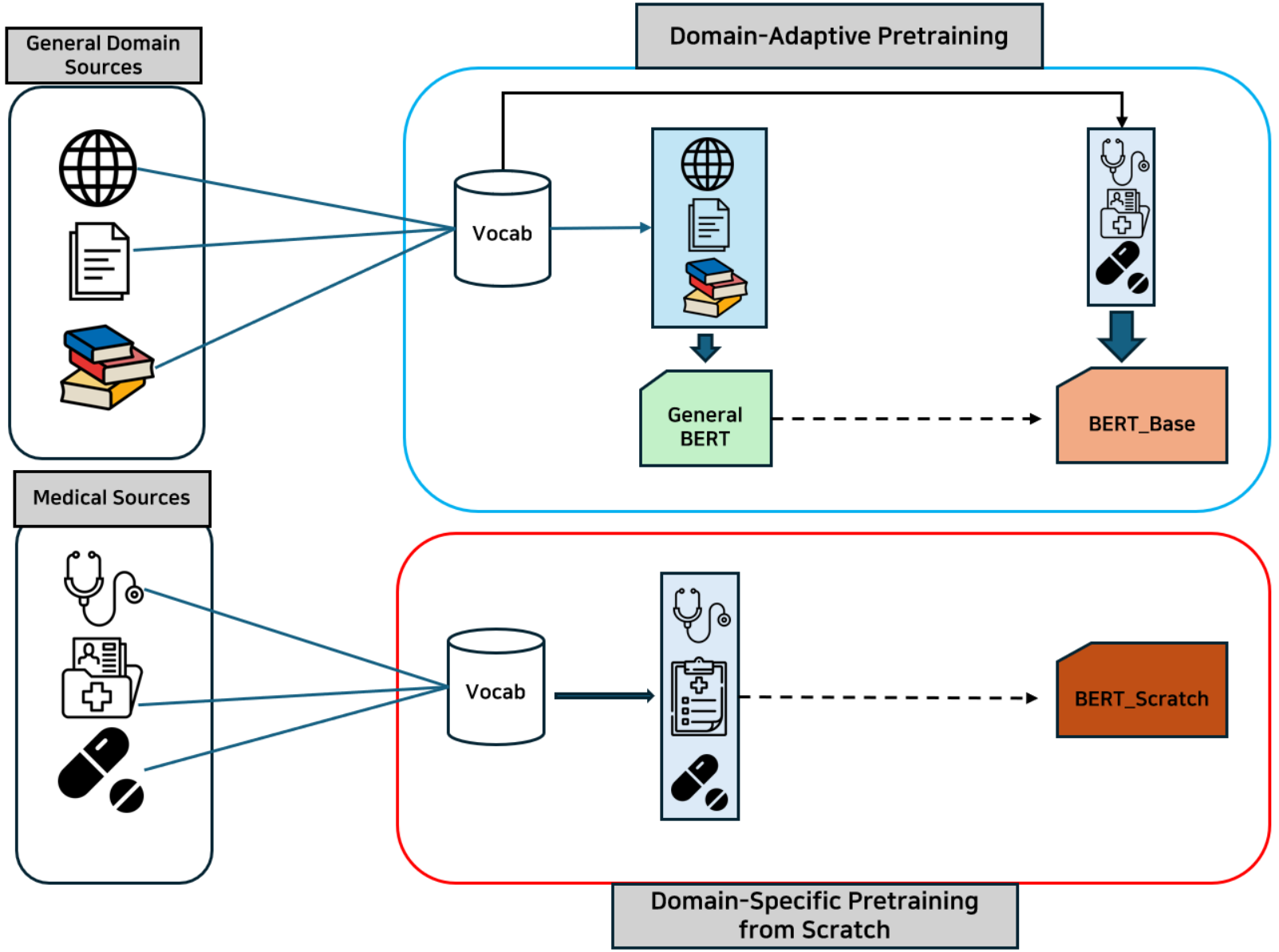
Backgrounds

BERT의 사전학습

BERT는 라벨 없는 대량의 텍스트로 두 과제(MLM·NSP)를 동시에 수행한다.



Domain Adaptive Pretraining(DAP) vs From-Scratch pretraining(FSP)



Bioinformatics 2020 : BioBERT

BioBERT: a pre-trained biomedical language representation model for biomedical text mining

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Associate Editor: Jonathan Wren

Received on May 16, 2019; revised on July 29, 2019; editorial decision on August 25, 2019; accepted on September 5, 2019

Abstract

Motivation: Biomedical text mining is becoming increasingly important as the number of biomedical documents rapidly grows. With the progress in natural language processing (NLP), extracting valuable information from biomedical literature has gained popularity among researchers, and deep learning has boosted the development of effective biomedical text mining models. However, directly applying the advancements in NLP to biomedical text mining often yields unsatisfactory results due to a word distribution shift from general domain corpora to biomedical corpora. In this article, we investigate how the recently introduced pre-trained language model BERT can be adapted for biomedical corpora.

Results: We introduce BioBERT (Bidirectional Encoder Representations from Transformers for Biomedical Text Mining), which is a domain-specific language representation model pre-trained on large-scale biomedical corpora. With almost the same architecture across tasks, BioBERT largely outperforms BERT and previous state-of-the-art models in a variety of biomedical text mining tasks when pre-trained on biomedical corpora. While BERT obtains performance comparable to that of previous state-of-the-art models, BioBERT significantly outperforms them on the following three representative biomedical text mining tasks: biomedical named entity recognition (0.62% F1 score improvement), biomedical relation extraction (2.80% F1 score improvement) and biomedical question answering (12.24% MRR improvement). Our analysis results show that pre-training BERT on biomedical corpora helps it to understand complex biomedical texts.

Availability and implementation: We make the pre-trained weights of BioBERT freely available at https://github.com/naver/biobert-pretrained, and the source code for fine-tuning BioBERT available at https://github.com/dmis-lab/biobert.

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Domain-Adaptive Pretraining

- 범용 도메인(English Wikipedia + BooksCorpus)으로 사전학습된 BERT를 생의학 말뭉치(PubMed 초록 4.5B + PMC 전문 13.5B 단어)로 추가 사전학습
- 기존 BERT의 범용 WordPiece 어휘를 그대로 승계 → 토크나이저가 일반 도메인에 최적화된 상태로 고정

2 Approach

In this article, we introduce BioBERT, which is a pre-trained language representation model for the biomedical domain. The overall process of pre-training and fine-tuning BioBERT is illustrated in Figure 1. First, we initialize BioBERT with weights from BERT, which was pre-trained on general domain corpora (English Wikipedia and BooksCorpus). Then, BioBERT is pre-trained on biomedical domain corpora (PubMed abstracts and PMC full-text articles). To show the effectiveness of our approach in biomedical text mining, BioBERT is fine-tuned and evaluated on three popular biomedical text mining tasks (NER, RE and QA). We test various pre-training strategies with different combinations and sizes of general domain corpora and biomedical corpora, and analyze the effect of each corpus on pre-training. We also provide in-depth analyses of BERT and BioBERT to show the necessity of our pre-training strategies.

ACM 2021 : PubMedBERT

Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing

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Pretraining large neural language models, such as BERT, has led to impressive gains on many natural language processing (NLP) tasks. However, most pretraining efforts focus on general domain corpora, such as newswire and Web. A prevailing assumption is that even domain-specific pretraining can benefit by starting from general-domain language models. In this paper, we challenge this assumption by showing that for domains with abundant unlabeled text, such as biomedicine, pretraining language models from scratch results in substantial gains over continual pretraining of general-domain language models. To facilitate this investigation, we compile a comprehensive biomedical NLP benchmark from publicly-available datasets. Our experiments show that domain-specific pretraining serves as a solid foundation for a wide range of biomedical NLP tasks, leading to new state-of-the-art results across the board. Further, in conducting a thorough evaluation of modeling choices, both for pretraining and task-specific fine-tuning, we discover that some common practices are unnecessary with BERT models, such as using complex tagging schemes in named entity recognition (NER). To help accelerate research in biomedical NLP, we have released our state-of-the-art pretrained and task-specific models for the community, and created a leaderboard featuring our BLURB benchmark (short for Biomedical Language Understanding & Reasoning Benchmark) at https://aka.ms/BLURB.

From-Scratch Pretraining

- 범용 도메인 사전학습 없이 생의학 말뭉치(PubMed 초록 약 1,400만 건 / 31억 단어)만으로 어휘와 모델 가중치를 처음부터 학습
- 의료 도메인 전용 어휘를 직접 구축 → 의학 용어(naloxone, lidocaine 등)를 과도하게 분절하지 않고 의미 단위로 보존

Biomedical Term	Category	BERT	SciBERT	PubMedBERT (Ours)
diabetes	disease	✓	✓	✓
leukemia	disease	✓	✓	✓
lithium	drug	✓	✓	✓
insulin	drug	✓	✓	✓
DNA	gene	✓	✓	✓
promoter	gene	✓	✓	✓
hypertension	disease	hyper-tension	✓	✓
nephropathy	disease	ne-ph-rop-athy	✓	✓
lymphoma	disease	l-ym-ph-oma	✓	✓
lidocaine	drug	lid-oca-ine]	✓	✓
oropharyngeal	organ	oro-pha-ryn-ge-al	or-opharyngeal	✓
cardiomyocyte	cell	card-iom-yo-cy-te	cardiomy-ocyte	✓
chloramphenicol	drug	ch-lor-amp-hen-ico-l	chlor-amp-hen-icol	✓
RecA	gene	Rec-A	Rec-A	✓
acetyltransferase	gene	ace-ty-lt-ran-sf-eras-e	acetyl-transferase	✓
clonidine	drug	cl-oni-dine	clon-idine	✓
naloxone	drug	na-lo-xon-e	nal-oxo-ne	✓

Table 1. Comparison of common biomedical terms in vocabularies used by the standard BERT, SciBERT and PubMedBERT (ours). A ✓ indicates the biomedical term appears in the corresponding vocabulary, otherwise the term will be broken into word pieces as separated by hyphen. These word pieces often have no biomedical relevance and may hinder learning in downstream tasks.

Scientific Reports 2022 : KM-BERT

A pre-trained BERT for Korean medical natural language processing

Yoojoong Kim[1,8], Jong-Ho Kim[2,5,8], Jeong Moon Lee[2], Moon Joung Jang[2], Yun Jin Yum[2,3], Seongtae Kim[4], Unsub Shin[4], Young-Min Kim[7], Hyung Joon Joo[2,5,6,9]✉ & Sanghoun Song[4,9]✉

With advances in deep learning and natural language processing (NLP), the analysis of medical texts is becoming increasingly important. Nonetheless, despite the importance of processing medical texts, no research on Korean medical-specific language models has been conducted. The Korean medical text is highly difficult to analyze because of the agglutinative characteristics of the language, as well as the complex terminologies in the medical domain. To solve this problem, we collected a Korean medical corpus and used it to train the language models. In this paper, we present a Korean medical language model based on deep learning NLP. The model was trained using the pre-training framework of BERT for the medical context based on a state-of-the-art Korean language model. The pre-trained model showed increased accuracies of 0.147 and 0.148 for the masked language model with next sentence prediction. In the intrinsic evaluation, the next sentence prediction accuracy improved by 0.258, which is a remarkable enhancement. In addition, the extrinsic evaluation of Korean medical semantic textual similarity data showed a 0.046 increase in the Pearson correlation, and the evaluation for the Korean medical named entity recognition showed a 0.053 increase in the F1-score.

In the field of clinical NLP, the medical history of patients, diagnosis, and treatment information are crucial for text processing. Therefore, word-level language representation models such as BioWordVec[1] have been developed and studied using Word2Vec[2] and Fasttext[3]. However, medical text is extremely challenging to analyze because of the complexity of word representations and jargon. Therefore, the adoption of a domain-specific approach is required for the development of deep-learning-based language models in medical text processing.

With significant progress in deep learning over the past few years, natural language processing (NLP) models have been developed to overcome the limitations of existing models used for language processing. Bidirectional encoder representations from transformers (BERT) is a deep learning language model constructed using the encoder structure from the transformer[4]. BERT has shown impressive performance in various NLP tasks and has revolutionized the NLP domain. Numerous language models are being developed using BERT[5–7].

BERT is a pre-trained language model that focuses on English. Furthermore, a multi-linguistic version of BERT architecture can be developed by modifying the vocabulary of the model and using relevant data used for training[4]. However, this approach does not sufficiently reflect various properties of a language[8]. Therefore, language-specific BERT has been studied in various languages. The Korean-based BERT (KR-BERT) is a representative language-specific model of the Korean language[9]. This model has shown remarkable results and excellent performance compared with other models trained using the Korean corpus. However, the model still has a chance to improve language understanding for text analysis in the specialized domain when the existing domain-specific English models are considered[10–13]. Domain-specific BERT models have been studied in the medical, financial, and legal fields. In particular, BioBERT showed a performance increase compared to the existing BERT model by training the model with the biomedical corpus[10]. This demonstrates the requirement for domain-specific training for BERT.

Domain-Adaptive Pretraining

- 일반 도메인에서 학습된 KR-BERT를 한국어-의료 말뭉치(의학 교과서·건강정보 뉴스·의학 논문, 약 6M/ 116M 단어)로 추가 사전학습
- 토크나이저를 그대로 사용 → 의학 용어 과분절 문제 잔존 (예: ibuprofen → i / ##bu / ##pro / ##fen)
- 희소 단어 문제를 해결하기 위해 의학 용어 10,562개를 어휘집에 추가(KM-BERT-vocab)했으나 개선 X
- 처음부터 사전학습을 수행하였을 경우 보다 유의미한 결과를 기대할 수 있다고 기술함

Model	Pearson correlation	Spearman correlation
KM-BERT	0.869	0.860
KM-BERT-vocab	0.851	0.834
KR-BERT	0.823	0.811
M-BERT	0.842	0.830

Table 3. Extrinsic evaluation results on MedSTS.

선행 연구 비교

	Domain-Adaptive Pretraining (DAP)	From-Scratch Pretraining (FSP)
영어-의료 도메인	BioBERT	PubMedBERT
한국어-의료 도메인	KM-BERT	★

DAP의 한계
- 일반 도메인으로 사전학습된 BERT는 추가 사전학습(DAP)만으로 도메인에 최적이 아닌 초기화를 완전히 되돌리기 어려움 [1]
- 이는 새로운 어휘를 추가하는것만으로 해결하기 어려울 수 있음 [2]

FSP의 차별점
- 토크나이저·임베딩을 도메인에 맞춰 한 번에 최적화 → 다운스트림 태스크에서 DAP를 상회함이 입증됨 [1]

연구 공백
- 한국어-의료 도메인에서 FSP를 구축한 사례는 아직 보고된 바 없음

[1] PubMedBERT : Gu et al., *ACM Trans. Comput. Healthcare*, 2021
[2] KM-BERT : Kim et al., *Sci. Rep.*, 2022

Methods - pretraining

1. 데이터셋 구성
2. 토크나이저 구성
3. 실험 설계

데이터셋 구성

출처	문서 수	단어 수
위키피디아·나무위키	21,025	3,576,818
AI-Hub	14,163	8,115,949
KCI 초록	42,215	5,855,588
MSD	3,399	1,971,990
국립국어원 신문 말뭉치	376,415	110,839,624
합계	499,432	136,215,557

- 위키피디아·나무위키의 의학/약학 문서 – 덤프 활용
- AI-Hub 의학지식 데이터 - 공개 데이터 수집
- KCI 의학·약학 초록 - 공식 API 수집
- MSD 매뉴얼 - 웹 크롤링
- 국립국어원 신문 말뭉치(건강·미용) - 배포 자료 활용
- 총 499,432 문서 / 약 136M 단어

토크나이저 구성

- 범용 토크나이저(DAP) — KLUE-BERT 어휘 사용
- 도메인 특화 토크나이저(FSP) — KCI 초록으로 학습

예제 문장

심근경색 후 발생한 심막염에 methylprednisolone을 정맥 투여하였다.

범용 토크나이저 (DAP) — 27 tokens

[CLS] 심근 ##경 ##색 후 발생 ##한 심 ##막염 ##에 me ##th ##yl ##pr ##ed
##n ##is ##ol ##one ##을 정맥 투여 ##하 ##였 ##다 . [SEP]

도메인 특화 토크나이저 (FSP) — 12 tokens

[CLS] 심근경색 후 발생한 심막 ##염에 methylprednisolone ##을 정맥 투여하였다 .
[SEP]

→ 도메인 특화 토크나이저가 같은 말뭉치를 33.5% 적은 토큰으로 표현

실험 설계 - 프리트레인

모델 구성

두 모델 모두 BERT-base 구조 + MLM + NSP로 사전학습
BERT_Base (DAP): 일반 도메인 가중치 → 의료 말뭉치 추가 학습
BERT_Scratch (FSP): 의료 말뭉치만으로 처음부터 학습

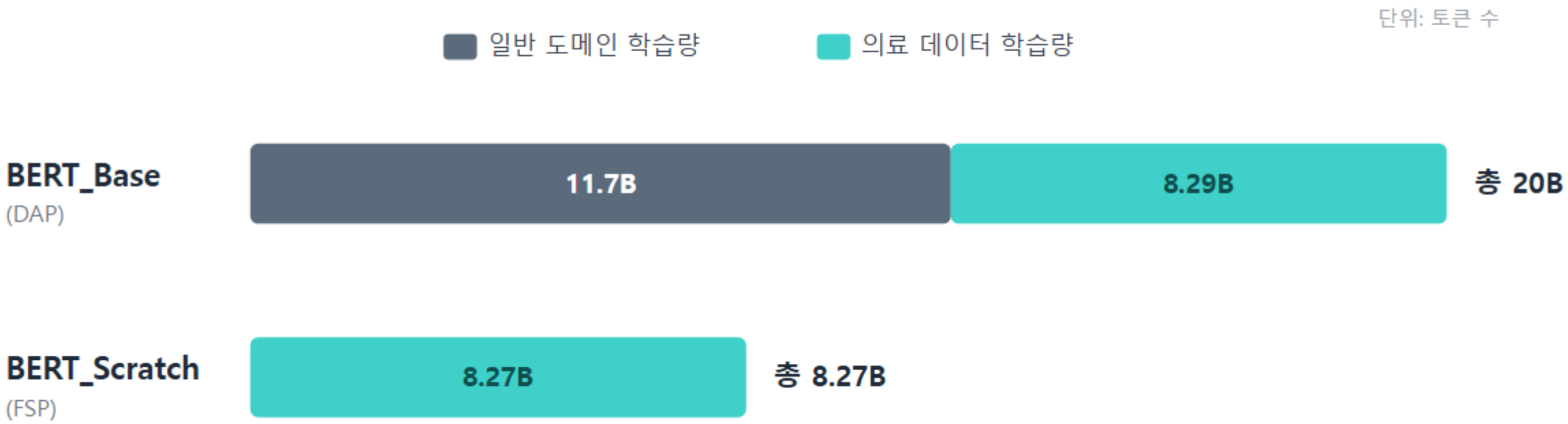
토큰 수 통제

토크나이저별 토큰 수 차이 → 의료 데이터 학습량을 약 8.28B 토큰 수준으로 일치

하이퍼파라미터

항목	BERT_Base (DAP)	BERT_Scratch (FSP)
초기 가중치	일반 도메인 사전학습	랜덤 초기화
토크나이저	KLUE-BERT	KCI 초록 학습
Epochs	20	30
의료 데이터 학습량	**8.29B**	**8.27B**
Learning rate	1e-4	1e-4

총 학습 토큰 수



BERT_Base는 한국어 위키피디아·나무위키로 처음부터 자체 사전학습

→ 의료 데이터 학습량은 같으나(약 8.28B) 총 학습량은 FSP가 약 2.4배 적음

Methods - finetuning

1. 데이터셋 소개
2. 실험 설계

데이터셋 소개

KorMedQA

의사·간호사·약사·치과의사 국가시험 기반 5지선다 의료 QA

Q. 25세 남자가 헌혈을 하고자 한다. 「혈액관리법」에 따라 채혈 전 시행해야 하는 건강진단 항목은?

1. 소변검사
2. 가슴 X선
3. **혈압 측정**
4. 심전도검사
5. 컴퓨터단층촬영

→ 정답: **3** (혈압 측정)

보기 5개 중 정답 1개 선택

KBMC

임상 문장의 각 어절에 의료 개체명 태그 부여 (NER)

질환(Disease) 신체(Body) 치료(Treatment) 기타(O)

심근경색	후	발생한	심장	주변	심막염에
Disease	O	O	Body	O	Disease

methylprednisolone을	정맥	투여하였다
Treatment	O	O

어절마다 4개 태그 중 하나로 분류

실험 설계 - 파인튜닝

파인튜닝 설정 (DAP·FSP 공통)

항목	KorMedQA	KBMC
과제 유형	다중 선택 (5지선다 QA)	토큰 분류(NER)
Epochs	20	5
Learning rate	5e-5	1e-4
Batch size	64	64
평가 지표	**Accuracy**	**Macro F1**

- 두 모델 모두 동일한 설정으로 파인튜닝
- 동일 epoch 적용 — 토크나이저 차이에 따른 별도 보정 없음

Results

1. 결과 소개
2. 결론

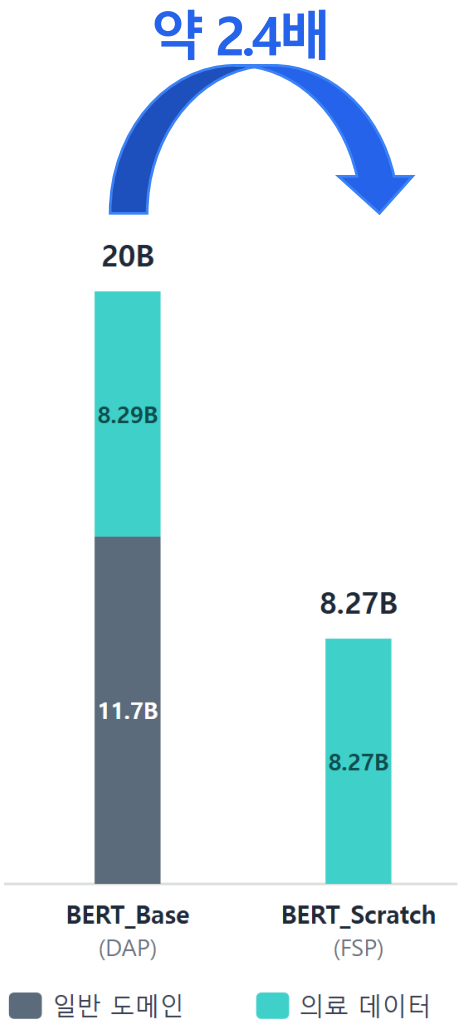
결과 소개

KorMedQA

	Doctor	Nurse	Pharm	Dentist	Total
BERT_Base	28.74	31.44	**23.50**	20.59	25.79
BERT_Scratch	**29.89**	**31.89**	**23.50**	**24.54**	**27.15**

KBMC

	Disease	Body	Treatment	Macro F1
BERT_Base	**89.85**	81.18	74.18	81.74
BERT_Scratch	89.35	**82.58**	**76.15**	**82.69**



결론

- 총 학습량이 2.4배 적은 FSP가 두 과제 종합 성능에서 DAP를 능가 (KorMedQA **+1.36%p**, KBMC **+0.95%p**)
- 도메인 특화 토크나이저가 의학 용어의 과분절을 해소하여 어휘·가중치를 처음부터 도메인에 최적화된 방향으로 사전학습한 결과
- 제한된 한국어-의료 말뭉치 환경에서 FSP와 DAP를 동일 조건으로 비교한 시도

→ **한국어-의료 도메인에서 FSP의 필요성을 시사**



Discussion

1. Limitations
2. Future Work

Limitations

한계

- 대규모 한국어 임상 말뭉치의 부재로, 그동안 한국어 의료 FSP 연구가 시도되기 어려웠음
- 본 연구 또한 임상 텍스트가 부족하여 신문의 건강·미용 기사를 선별·보완해 사용 → 순수 임상 도메인과는 차이
- GPU 자원의 제약으로 사전학습 시간이 충분하지 않아, 총 학습량이 타 모델에 비해 적은 수준

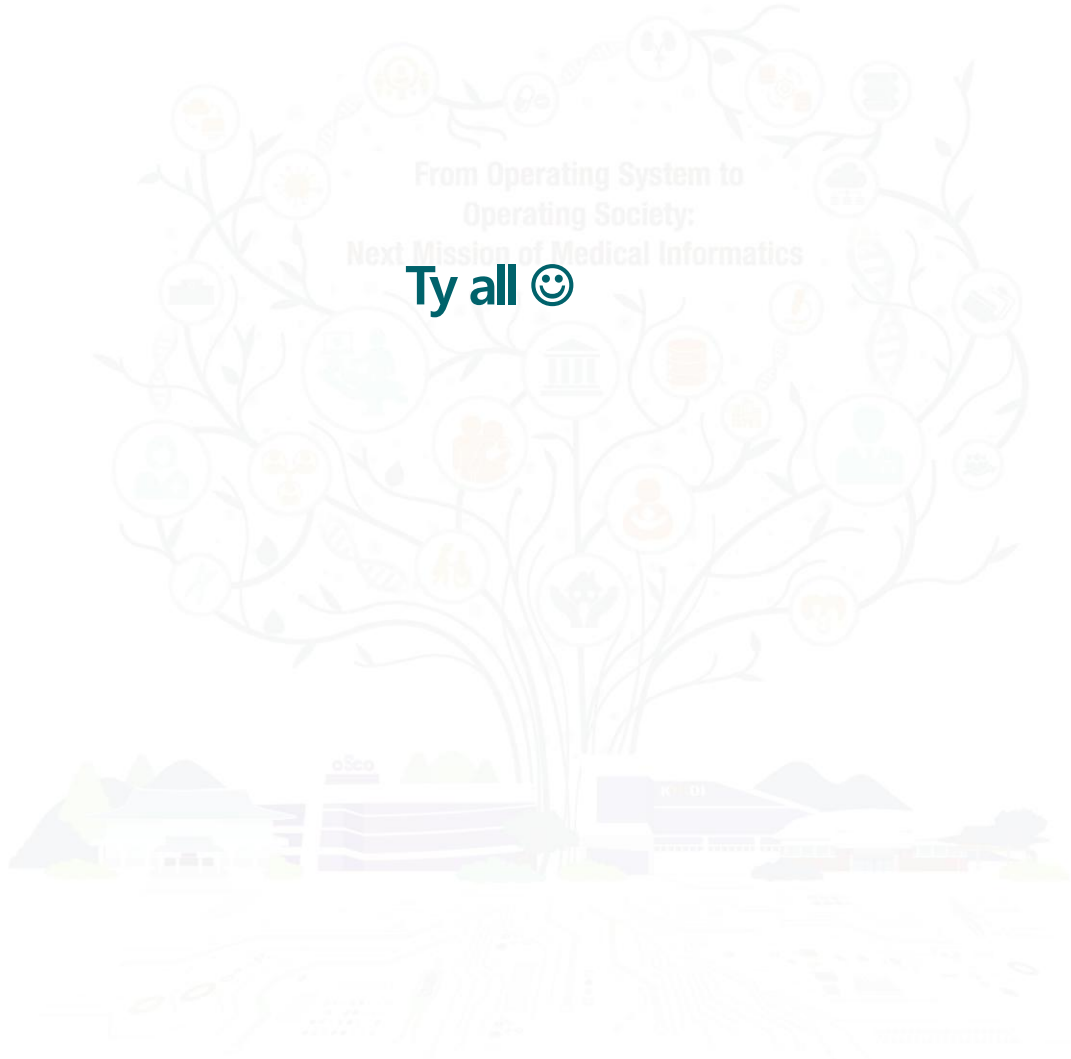
Future Work

향후 연구 방향

- K-MIMIC 등 한국어 대규모 임상 데이터가 공개되면 순수 임상 데이터만으로 모델 구축 가능
- 본 연구의 FSP - DAP 비교를 더 큰 임상 말뭉치로 확장해 유효성을 검증
- BERT 기반 인코더를 넘어, 생성형 디코더 모델(의료 LLM)에 FSP를 적용해 성능·활용 범위 확대


From Operating System to
Operating Society:
Next Mission of Medical Informatics

Q & A
AMA


From Operating System to
Operating Society:
Next Mission of Medical Informatics

Ty all ☺